

Requirement Analysis – JobAlign

Business Problem:

Job postings are often written in different formats and use inconsistent terms for similar skills. This makes it difficult for job seekers to quickly understand job requirements and determine whether their skills match a particular position. Candidates also lack a structured way to identify missing skills, understand their strengths, and receive personalized feedback for improving their employability.

Business Goal:

Build a centralized platform that:

- Converts unstructured job postings into standardized information.
- Normalizes different names for the same skills.
- Compares job requirements with candidate skills.
- Calculates a job-fit score.
- Identifies skill gaps.
- Provides personalized AI-powered feedback.
- Helps candidates understand what skills they should improve.

Stakeholders:

Stakeholder	Interest
Job Seekers	Analyze jobs and understand their skill compatibility.
Recruiters	Create and manage standardized job postings.
Administrators	Manage users, skills, aliases and system configuration.
Organizations	Improve consistency and quality of job postings.

Actors:

1. Candidate
2. Recruiter
3. Administrator
4. AI Service – External system used for information extraction, summarization and feedback.

Functional Requirements:

Priority: M – Must Have | S – Should Have | C – Could Have

1. Candidate:

ID	Requirement	Priority
FR-01	Register, log in, and manage profile details.	M
FR-02	Add, update, and remove skills, experience, education, projects, and certifications.	M
FR-03	Upload and analyze a resume.	M
FR-04	Save and analyze job postings.	M
FR-05	View normalized job details, required and preferred skills.	M
FR-06	View job-fit score, matched skills, and skill gaps.	M
FR-07	Receive AI-generated feedback and skill improvement suggestions.	M
FR-08	Compare multiple job postings and view analysis history.	S

2. Recruiter:

ID	Requirement	Priority
FR-09	Create, edit, and manage job postings.	M
FR-10	View normalized job information and extracted skills.	M
FR-11	Review job posting completeness and edit extracted details.	S
FR-12	Manage required and preferred skills.	M

3. Administrator:

ID	Requirement	Priority
FR-13	Manage users, roles, and access.	M
FR-14	Manage skills, skill aliases, and categories.	M
FR-15	Manage duplicate or inappropriate job postings.	S
FR-16	View system reports and skill-gap statistics.	C

4. AI Services:

ID	Requirement	Priority
FR-17	Extract job details and skills from unstructured postings.	M
FR-18	Extract relevant information from resumes.	M
FR-19	Generate concise job summaries.	S
FR-20	Generate personalized skill-gap feedback and recommendations.	M

Non-Functional Requirements:

ID	Category	Requirement
NFR-01	Performance	Job analysis should complete within 10 seconds for most submissions.
NFR-02	Performance	The saved-jobs list should load within 2 seconds for up to 500 postings.
NFR-03	Security	All protected features shall require authentication and role-based authorization.
NFR-04	Security	Passwords shall be securely hashed and data shall be encrypted during transmission.
NFR-05	Reliability	Failure of the AI service shall not prevent a job posting from being saved.
NFR-06	Accuracy	Extracted details should include confidence indicators, and low-confidence results should be flagged for user review.
NFR-07	Auditability	Original job-posting text and relevant analysis information shall be retained for traceability.
NFR-08	Privacy	Only necessary resume and job-posting data shall be shared with external AI services, and users shall be able to delete their data.
NFR-09	Usability	The application shall provide a simple, clear and responsive interface for desktop and mobile screens.
NFR-10	Scalability	The system shall support increasing numbers of users, resumes and job postings without major redesign.
NFR-11	Maintainability	The system shall use modular components so that individual features can be modified independently.
NFR-12	Availability	The application shall remain accessible and operational under normal usage conditions.

User Stories:

- As a candidate, I want to create my skill profile and upload my resume so that I can compare my qualifications with job requirements.
- As a candidate, I want to analyze and compare job postings so that I can find suitable opportunities.
- As a candidate, I want to see my job-fit score, skill gaps, and AI feedback so that I know how to improve.
- As a recruiter, I want to create and manage job postings so that candidates can understand the requirements.

- As a recruiter, I want job postings to be normalized and checked for completeness so that information is presented consistently.
- As an administrator, I want to manage users, roles, skills, and aliases so that the system remains secure and consistent.
- As an administrator, I want to manage job postings and view skill-gap reports so that I can maintain data quality and monitor trends.

Given – When – Then:

Given the candidate is logged in and has a completed skill profile,

When the candidate submits a valid job description,

Then the system should extract and normalize the job requirements, compare them with the candidate's skills, calculate a match score, identify skill gaps, and generate personalized feedback.

Acceptance Criteria – Job Analysis:

As a candidate, I want to analyze a job posting so that I can understand its requirements and compare them with my skills.

#	Acceptance Criteria
1	The candidate must be logged in.
2	The candidate must provide a valid job description.
3	The system should extract the job title and relevant requirements.
4	The system should identify required and preferred skills.
5	The system should normalize recognized skills.
6	The system should compare the job requirements with the candidate profile.
7	The system should calculate a job-fit score.
8	The system should display matched and missing skills.
9	The system should provide personalized feedback.
10	The candidate should be able to view the completed analysis.

Example of Skill Gap Feedback:

Job Requirements:

- C#
- ASP.NET Core
- SQL Server
- REST API

- Docker
- Azure

Candidate Skills:

- C#
- ASP.NET Core
- SQL Server
- REST API

System Result:

Match Score: 80%

Matched Skills:

- C#
- ASP.NET Core
- SQL Server
- REST API

Missing Skills:

- Docker
- Azure

Feedback:

You have a strong foundation for this position and meet the major backend development requirements. Docker and Azure are the main skill gaps. Prioritizing these technologies would improve your suitability for similar .NET roles.

Main System Modules:

1. Authentication & User Management
2. Candidate Profile Management
3. Job Posting Management
4. AI Job Information Extraction
5. Job Posting Normalization
6. Skill & Skill Alias Management
7. Resume Analysis
8. Job Matching
9. Skill Gap Analysis
10. AI Feedback Generation
11. Job Comparison
12. Dashboard & Reports

From Requirement-to-Development:

Stage	JobAlign Example
Business Requirement	Candidate should know how well their skills match a job.
User Story	Candidate wants to analyze a job and check suitability.
Acceptance Criteria	Job is analyzed, skills are matched, and score and gaps are shown.
System Design	MVC, Web API, Matching Service, and AI Service.
Database Design	Users, Jobs, Skills, JobSkills, UserSkills, Matches, and Feedback.
Development Task	Build job analysis, skill matching, and feedback features.
Sprint	Develop and test the feature in the planned sprint.

Future Scope:

- Integration with external job portals
- Automated job recommendations
- Personalized learning-resource recommendations
- Advanced semantic job matching
- Mobile application
- Recruiter-side candidate ranking

Expected Outcome:

JobAlign will provide candidates with a structured way to understand job requirements, measure their compatibility with a position, identify missing skills, and receive actionable feedback for improvement.

The system will transform the process from simply **"finding a job"** into **"understanding how well your skills fit the job and what you can improve."**